

# Esteban Carisimo

🏠 [estcarisimo.github.io](https://estcarisimo.github.io) | [@estcarisimo](#) | [Esteban Carisimo](#)

## Personal Profile

Systems Engineer at Cloudflare on the Protocols team, applying statistical tools to improve network performance. My research background focuses on Internet topology, particularly on performance, resilience, and the role of governments in shaping the Internet. I have studied key Internet infrastructures such as Submarine Cable Networks (SCNs), Content Delivery Networks (CDNs), Internet Exchange Points (IXPs), and Public DNS resolvers, and explored how governments influence Internet development, accessibility, and control.

## Education

### Universidad de Buenos Aires

Buenos Aires, Argentina

Doctor of Engineering

Apr 2015 - Apr 2020

- Advisor: J.I. Alvarez-Hamelin (UBA); Co-Advisor: Amogh Dhamdhere (UC San Diego)
- Dissertation: *Internet tomography: Traffic measurement and its relationship with topology*
- Summa Cum Laude

### Universidad de Buenos Aires

Buenos Aires, Argentina

Engineering Degree in Electronics

Mar 2009 - Dec 2014

- Diploma de Honor (*Honor for outstanding GPA*)
- Dissertation: *Network tomography: Measurements on an Internet Exchange Point*

## Appointments

### Cloudflare

Austin, TX, USA

Systems Engineer

Jan 2026 - Present

- Protocols Team — applying statistical tools to improve network performance

### Department of Computer Science, Northwestern University

Evanston, IL, USA

Postdoctoral Fellow

Apr 2021 - Dec 2025

- Mentor: Fabian E. Bustamante

### Department of Computer Science, School of Engineering, Universidad de Buenos Aires

Buenos Aires, Argentina

Assistant Professor (Spanish: Profesor Regular Adjunto)

Aug 2019 - Apr 2021

- Tenured Teaching Appointment
- Instructor of 75.43 - Introduction to Distributed Systems

### Department of Computer Science, School of Engineering, Universidad de Buenos Aires

Buenos Aires, Argentina

Head of Teaching Assistants

Mar 2017 - Jul 2019

- 75.43 - Introduction to Distributed Systems

### CONICET (Argentina's National Scientific and Technical Research Council)

Buenos Aires, Argentina

Doctoral Fellow

Apr 2015 - Apr 2020

- Mentor: J.I. Alvarez-Hamelin

### CAIDA, San Diego Supercomputer Center, UC San Diego

La Jolla, CA, USA

Visiting Graduate Student

Oct 2017 - Apr 2018

- Researched automatic detection of persistent congestion in the Internet.
- Mentor: Amogh Dhamdhere

## Awards

2022 **Best Dataset Award**, *Quantifying Nations' Exposure to Traffic Observation and Selective Tampering*. Passive and Active Measurement (PAM) Conference

Virtual Event

† indicates mentee

## JOURNAL ARTICLES

- [J4] Take The Long Way Home - Distant Peering to the Cloud  
**Esteban Carisimo**, Mia Weaver, Fabián E. Bustamante, Paul Barford  
*IEEE ToN* pp. 398–410. 2026
- [J3] Network topology facilitates internet traffic control in autocracies.  
Eda Keremoğlu, Nils B Weidmann, Alexander Gamero-Garrido, **Esteban Carisimo**, Alberto Dainotti, Alex C Snoeren  
*PNAS Nexus*. pgae069. 2024
- [J2] A First Look at the Latin American IXPs.  
**Esteban Carisimo**, Julián M. Del Fiore, Diego Dujovne, Cristel Pelsser, J. Ignacio Alvarez-Hamelin  
*SIGCOMM Comput. Commun. Rev.* pp. 18–24. 2020
- [J1] Studying the evolution of content providers in IPv4 and IPv6 internet cores.  
**Esteban Carisimo**, Carlos Selmo, J. Ignacio Alvarez-Hamelin, Amogh Dhamdhere  
*Computer Communications*. pp. 54–65. 2019

## CONFERENCE PROCEEDINGS

- [C16] Calypso's Voyage: Charting Traceroute Paths to Submarine Cables  
Caleb J. Wang†, Ying Zhang†, Qianli Dong†, **Esteban Carisimo**, Ramakrishnan Durairajan, Fabián E. Bustamante  
*ACM IMC*, 2026
- [C15] En Unión y Libertad: Subnational Strategies for Hosting Government Services  
**Esteban Carisimo**, Mariano G. Beiró, Lukas De Angelis Riva†, Mauricio Buzzzone†, Fabián E. Bustamante  
*Passive and Active Measurement*, 2026, Acceptance rate 31.57% (18/57)
- [C14] Monitoring Latency on Submarine Cables: Limitations and Opportunities  
Mia Weaver, Darryl Veitch, Paul Barford, Fabián E. Bustamante, **Esteban Carisimo**,  
*Proc. of NINES*, 2026, Acceptance rate 48.27% (28/58)
- [C13] The Aleph: Decoding Geographic Information from DNS PTR Records Using Large Language Models  
Kedar Thiagarajan†, **Esteban Carisimo**, Fabián E. Bustamante  
*ACM CoNEXT*, 2025
- [C12] Learning AS-to-Organization Mappings with Borges  
Carlos Selmo, **Esteban Carisimo**, Fabián E. Bustamante, J. Ignacio Alvarez-Hamelin  
*ACM IMC*, 2025, Acceptance rate 17.44% (15/86)
- [C11] DarkSim: A similarity-based time-series analytic framework for darknet traffic  
Max Gao†, Ricky P. K. Mok, **Esteban Carisimo**, Eric Li, Shubham Kulkarni, k Claffy  
*ACM IMC*, 2024, Acceptance rate 21.34% (54/253)
- [C10] Of Choices and Control - A Comparative Analysis of Government Hosting  
Rashna Kumar†, **Esteban Carisimo**, Lukas De Angelis Riva†, Mauricio Buzzzone†, Fabián E. Bustamante, Ihsan Ayyub Qazi, Mariano G. Beiró  
*ACM IMC*, 2024, Acceptance rate 21.34% (54/253)
- [C9] Ten years of the Venezuelan crisis - An Internet perspective  
**Esteban Carisimo**, Rashna Kumar†, Caleb J. Wang†, Santiago Klein†, Fabián E. Bustamante  
*ACM SIGCOMM*, 2024, Acceptance rate 16.94% (62/366)
- [C8] A hop away from everywhere: A view of the intercontinental long-haul infrastructure.  
**Esteban Carisimo**, Caleb J. Wang†, Mia Weaver, Fabián E. Bustamante, Paul Barford  
*Proc. ACM Meas. Anal. Comput. Syst.* 2023, Acceptance rate 12.50% (10/80)
- [C7] Destination Unreachable: Characterizing Internet Outages and Shutdowns.  
Zachary Bischof, Kennedy Pitcher, **Esteban Carisimo**, Amanda Meng, Rafael Nunes, Ramakrishna Padmanabhan, Margaret E. Roberts, Alex C. Snoeren, Alberto Dainotti  
*ACM SIGCOMM*, 2023, Acceptance rate 22.46% (73/325)
- [C6] as2org+: Enriching AS-to-Organization Mappings with PeeringDB.  
Augusto Arturi†, **Esteban Carisimo**, Fabián E. Bustamante  
*Passive and Active Measurement*, 2023, Acceptance rate 33.75% (27/80)
- [C5] Jitterbug: A new framework for jitter-based congestion inference.  
**Esteban Carisimo**, Ricky K. P. Mok, David D. Clark, Kc Claffy  
*Passive and Active Measurement*, 2022, Acceptance rate 48.39% (30/62)
- [C4] Quantifying Nations' Exposure to Traffic Observation and Selective Tampering.  
Alexander Gamero-Garrido, **Esteban Carisimo**, Shuai Hao, Bradley Huffaker, Alex C. Snoeren, Alberto Dainotti  
*Passive and Active Measurement*, 2022, Acceptance rate 48.39% (30/62)
- [C3] Identifying ASes of State-Owned Internet Operators.  
**Esteban Carisimo**, Alexander Gamero-Garrido, Alex C. Snoeren, Alberto Dainotti  
*ACM Internet Measurement Conference (IMC)*, 2021, Acceptance rate 27.55% (54/196)
- [C2] Studying the Evolution of Content Providers in the Internet Core.

**Esteban Carisimo**, Carlos Selmo, J. Ignacio Alvarez-Hamelin, Amogh Dhamdhere  
*Network Traffic Measurement and Analysis Conference (TMA)*, 2018

[C1] Hidden Internet Topologies Info: Truth or Myth?  
Sofia Silva Berenguer, **Esteban Carisimo**, J. Ignacio Alvarez-Hamelin, Francisco Valera Pintor  
*Workshop on Fostering Latin-American Research in Data Communication Networks*, 2016

PREPRINTS

[R1] Linnaeus: A Hierarchical, Multi-Label Framework for Autonomous System Classification  
Marcos Piotto<sup>†</sup>, Ignacio Schuemer<sup>†</sup>, Santiago T. Torres<sup>†</sup>, Mariano G. Beiró, **Esteban Carisimo**, Fabián E. Bustamante  
*arXiv*, 2026

PEER-REVIEWED EXTENDED ABSTRACTS AND POSTERS

[P4] Threading the Ocean: Mapping Digital Routes Across Submarine Cables using Calypso  
Caleb J. Wang<sup>†</sup>, Ying Zhang<sup>†</sup>, Oscar Dong<sup>†</sup>, **Esteban Carisimo**, Ram Durairajan, Fabián E. Bustamante  
*Proc. of SIGCOMM Shorts*, 2025

[P3] A hop away from everywhere: A view of the intercontinental long-haul infrastructure  
**Esteban Carisimo**, Caleb J. Wang<sup>†</sup>, Mia Weaver, Fabián E. Bustamante, Paul Barford  
*Abstracts of ACM SIGMETRICS/IFIP PERFORMANCE*, 2024

[P2] Beyond Proximity: Exploring Remote Cloud Peering  
**Esteban Carisimo**, Mia Weaver, Fabián E. Bustamante, Paul Barford  
*Proc. of SIGCOMM Posters*, 2024

[P1] Revealing Hidden Secrets: Decoding DNS PTR records with Large Language Models  
Kedar Thiagarajan<sup>†</sup>, **Esteban Carisimo**, Fabián E. Bustamante  
*Proc. of SIGCOMM Posters*, 2024

Software, Systems and Datasets

WEB SERVICES

**The Aleph** PTR-based Geolocation method <https://thealeph.ai/>  
**NetCores** AS connectivity using k-core decomposition <https://netcores.fi.uba.ar>  
**LaNet-vi** Large Networks Visualization using k-core decomposition <https://lanet-vi.fi.uba.ar>

TOOLS

| TOOL                        | DESCRIPTION                                                          | SOURCE                                  |
|-----------------------------|----------------------------------------------------------------------|-----------------------------------------|
| <b>Borges</b>               | AI-based AS-to-Organizations mapping tool                            | PyPI: <code>borges</code>               |
| <b>Jitterbug</b>            | Network Congestion Inference tool                                    | PyPI: <code>jitterbug-inference</code>  |
| <b>LaNet-vi</b>             | Large Networks Visualization using k-core decomposition (Python)     | PyPI: <code>lanet-vi</code>             |
| <b>levy-stable</b>          | Lévy stable distributions fitting library (formerly pylevy)          | PyPI: <code>levy-stable</code>          |
| <b>as2org+</b>              | Regex-based AS-to-Organizations mapping tool                         | GitHub                                  |
| <b>bayesian-changepoint</b> | Online and offline Bayesian changepoint detection on PyTorch tensors | PyPI: <code>bayesian-changepoint</code> |
| <b>Smoking Pi</b>           | Continuous SmokePing-based network monitoring                        | apt, Homebrew: <code>smoking-pi</code>  |

DATASET

**State-Owned ASes** List of ASes of State-Owned Internet Providers <https://www.github.com/estcarisimo/state-owned-ases>

Talks

[T15] The Conflicted Relationship between Governments and the Internet  
**Esteban Carisimo**  
2025, Northeastern University, Boston, MA, USA

[T14] The Conflicted Relationship between Governments and the Internet  
**Esteban Carisimo**  
2024, Johns Hopkins University, Baltimore, MD, USA

[T13] The Conflicted Relationship between Governments and the Internet  
**Esteban Carisimo**  
2024, University of Maryland – College Park, College Park, MD, USA

[T12] Una decada de crisis en Venezuela: Una mirada desde la infraestructura de Internet  
**Esteban Carisimo**  
2024, LACNIC/LACNOG 2024, Asuncion, Paraguay

- [T11] Ten years of the Venezuelan crisis - An Internet perspective  
**Esteban Carisimo**  
2024, SIGCOMM 2024, Sydney, NSW, Australia
- [T10] A hop away from everywhere: A view of the intercontinental long-haul infrastructure  
**Esteban Carisimo**  
2024, AIMS-IYP UCSD, La Jolla, CA, USA
- [T9] A hop away from everywhere: A view of the intercontinental long-haul infrastructure  
**Esteban Carisimo**  
2024, SIGMETRICS, Venice, Italy
- [T8] El Dilema Digital de los Gobiernos: Entre Autonomia y Terceizacion  
**Esteban Carisimo**  
2024, Universidad de San Andres, San Fernando, Buenos Aires, Argentina
- [T7] Unexpected Peering Connections to the Cloud: Going the distance when you don't have to  
**Esteban Carisimo**  
2023, AIMS-IYP UCSD, La Jolla, CA, USA
- [T6] Unexpected Peering Connections to the Cloud: Going the distance when you don't have to  
**Esteban Carisimo**  
2023, University of Wisconsin-Madison, Madison, WI, USA
- [T5] as2org+: Enriching AS-to-Organization Mappings with PeeringDB  
**Esteban Carisimo**  
2023, PAM, Virtual Event
- [T4] Jitterbug: A new framework for jitter-based congestion inference  
**Esteban Carisimo**  
2022, PAM, Virtual Event
- [T3] Identifying ASes of State-Owned Internet Operators  
**Esteban Carisimo**  
2021, IMC, Virtual Event
- [T2] Studying the Evolution of Content Providers in the Internet Core  
**Esteban Carisimo**  
2018, LACNOG, Rosario, Argentina
- [T1] Studying the Evolution of Content Providers in the Internet Core  
**Esteban Carisimo**  
2018, TMA, Vienna, Austria

## Press and Broader Impact

---

- 2026 **Cloudflare Blog**, (Blog) When “idle” isn’t idle: how a Linux kernel optimization became a QUIC bug
- 2025 **ISOC Pulse**, (Blog) Mapping Who Really Runs the Internet: Introducing “Borges”
- 2024 **Academia Abierta**, (Newsletter) El dilema de los gobiernos en la era digital
- 2024 **ISOC Pulse**, (Blog) The Story of Venezuela’s Deteriorating Internet Infrastructure
- 2024 **Lawfare Daily**, (PODCAST) How Internet Infrastructure Affects Digital Repression in Venezuela
- 2024 **ISOC Pulse**, (Blog) A hop away from everywhere long-haul links in today’s Internet
- 2024 **APNIC Blog**, (Blog) A hop away from everywhere long-haul links in today’s Internet
- 2022 **APNIC Blog**, (Blog) Identifying Autonomous Systems of State-owned Internet operators

## Teaching

---

### CS340 - Introduction to Computer Networking

Northwestern University

Instructor

Jan 2022 - Dec 2024

- Terms:
  - Fall 2024 (Enrollment: 80, CTEC: 4.64 / 6.00)
  - Fall 2023 (Enrollment: 78, CTEC: 5.04/ 6.00)
  - Fall 2022 (Enrollment: 67, CTEC: 4.83/ 6.00)
  - Winter 2022 (Enrollment: 134, CTEC: 4.20 / 6.00)
- Website: <https://northwestern-cs340.github.io/>

### 75.43 - Introduction to Distributed Systems

Universidad de Buenos Aires

Instructor

Jul 2019 - Apr 2021

- Semesters: S2-2020, S1-2020, S2-2019
- Equivalent to Northwestern’s CS340 - Introduction to Computer Networking

## 75.43 - Introduction to Distributed Systems

Head of Teaching Assistants

- Semesters: S1-2019, S2-2018, S1-2018, S1-2017
- Equivalent to Northwestern's CS340 - Introduction to Computer Networking

Universidad de Buenos Aires

Mar 2017 - Jul 2019

## Service

---

### DEPARTMENT SERVICE

#### Latin CS

Faculty Advisor

- Student organization for self-identified as Latin American/ Latinx / Hispanic.
- Temporary invited to represent the faculty in mentoring the organization.

Northwestern University

Jan 2023 - Jun 2023

#### Admission Committee for the PhD in Computer Science

Committee Member

- Representing the System and Networking area.

Northwestern University

Jan 2022 - Jan 2023

#### Curriculum Committee of the CSE Undergraduate Program

Faculty Representative

- Co-governed board composed of faculty (3), graduate (2), and student (1) members

Universidad de Buenos Aires

Jun 2019 - Apr 2021

### CONFERENCE COMMITTEES AND JOURNAL REVIEWS

#### General Chair

- PAM 2024

#### Publicity Chair

- ACM IMC 2025

#### Program Committee Member

- 2027: ACM SIGMETRICS
- 2026: ACM IMC, ACM CoNEXT, PAM
- 2025: ACM IMC, ACM CoNEXT, PAM, ACM SIGMETRICS
- 2024: ACM IMC, TMA
- 2023: ACM IMC, PAM
- 2022: PAM, ACM IMC (Only Poster Session)

#### External Reviewer

- ACM/IEEE Transaction on Networking (ToN)
- ACM SIGCOMM Computer Communications Review (CCR)

## Mentoring

---

### PHD STUDENTS

1. **Rashna Kumar**. Mentor. Northwestern University. *Mar 2021 - Dec 2025*
2. **Caleb J. Wang**. Mentor. Northwestern University. *Sep 2023 - Dec 2025*
3. **Kedar Thiagarajan**. Mentor. Northwestern University. *Sep 2023 - Dec 2025*

### MASTER STUDENTS OR EQUIVALENT

1. **Bruno de Cruz Ricciardi**. Research Assistant Mentor (w/ Mariano G. Beiró). Universidad de San Andres. *Oct 2024 - Mar 2025*
2. **Santiago Torres**. Research Assistant Mentor (w/ Mariano G. Beiró). Universidad de San Andres. *Oct 2024 - Dec 2025*
3. **Marcos Piotto**. Research Assistant Mentor (w/ Mariano G. Beiró). Universidad de San Andres. *Oct 2024 - Dec 2025*
4. **Ignacio Schuemer**. Research Assistant Mentor (w/ Mariano G. Beiró). Universidad de San Andres. *Oct 2024 - Dec 2025*
5. **Mauricio Buzzzone**. Final Project Advisor. Universidad de Buenos Aires. *Oct 2023 - Nov 2024*
6. **Lukas De Angelis Riva**. Final Project Advisor. Universidad de Buenos Aires. *Oct 2023 - Nov 2024*
7. **Santiago Klein**. Final Project Advisor. Universidad de Buenos Aires. *Jan 2023 - Sep 2023*

8. **Nicolas Aguerre**. Final Project Advisor. Universidad de Buenos Aires. *Jan 2023 - Sep 2023*
9. **Raghul Ravi**. Final Project Advisor. Northwestern University. *Sep 2022 - Jun 2023*
10. **Aman Khalid**. Final Project Advisor. Northwestern University. *Jun 2022 - Dec 2022*
11. **Cecilia Hortas**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2021 - Dec 2022*
12. **Camila Bojman**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2021 - Dec 2022*
13. **Manuel Ruiz Echazu**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2021 - Aug 2022*
14. **Tomas Botalla**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2021 - Aug 2022*
15. **Augusto Arturi**. Master's Thesis Advisor. Universidad de Buenos Aires. *Sep 2020 - Aug 2021*
16. **Rodrigo Zapico**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2020 - Jun 2021*
17. **Martín Errázquin**. Final Project Advisor. Universidad de Buenos Aires. *Sep 2020 - Jun 2021*
18. **Sofía Morseletto**. Master's Thesis Advisor. Universidad de Buenos Aires (joint w University of Luxembourg). *Jan 2019 - Apr 2020*

## Languages

---

**English** Proficient  
**Spanish** Native  
**Italian** Fluent

## References

---

**References available upon request.**